SG3525A PWM Controller: Working, Pins, Duty Cycle and Frequency

1. What the SG3525A does

The SG3525A is a **PWM (pulse-width modulation) controller IC**. It does not handle power itself. It produces two square-wave signals (OUA and OUB) that switch a power transistor on and off. Two things about those signals matter:

- **Frequency**: how many pulses per second. It is set by a resistor and a capacitor (Rt, Ct, Rd).
- **Duty cycle**: the percentage of each period that the pulse is high. It is set by a control voltage.

2. How the IC works:

1. **Reference.** An internal regulator makes a steady 5.1 V (pin 16, VREF) from the supply. Every threshold in the circuit is derived from it, so the supply can change without upsetting the controller.
2. **Oscillator.** Rt sets how fast the capacitor Ct charges. Ct charges up to about 3.3 V and is then discharged quickly through Rd. This makes a **sawtooth** wave on pin 5 (CT) that ramps between roughly 0.9 V and 3.3 V. The sawtooth is the clock of the whole IC.
3. **Error amplifier.** An op amp compares the voltage on pin 2 (NI, non-inverting input) with pin 1 (INV, inverting input). Its output appears on pin 9 (CMP) and is the **control voltage**.
4. **PWM comparator.** It compares the sawtooth with the control voltage. While the sawtooth is below the control voltage, the pulse is high. When it rises above, the pulse is low.
5. **Steering.** A flip-flop (a toggle circuit) sends the pulses alternately to OUA and OUB. A logic gate keeps both outputs off during the Sawtooth's reset time (the dead time).
6. **Output stage.** Two totem-pole drivers (a pair of transistors that push the output to the supply or to ground) drive the gates of the power switches, powered from pin 13 (VC).

3. Significance of each pin:

Pin	Name	What it does
1	INV (inverting input)	One input of the error amplifier. Normally receives feedback, or is joined to CMP to make a follower.
2	NI (non-inverting input)	Other input of the error amplifier. Normally receives the reference or the set value.
3	SYNC	Lets the oscillator be synchronized to an external clock or to another IC. Not needed for a single IC.
4	OSC OUT	Output of the internal clock pulse. Used to synchronize other ICs.
5	CT	The timing capacitor. Its sawtooth voltage is compared against the control voltage.
6	RT	The timing resistor. It sets the charging current of Ct through an internal current mirror.
7	DISCHARGE	The discharge path for Ct. The resistor Rd between pin 5 and pin 7 sets the dead time.
8	SOFT-START	A capacitor here is charged slowly by an internal current source. The pulse width is held down until it charges, so the output rises gradually. This limits inrush current and overshoot.

Pin	Name	What it does
9	CMP (compensation)	Output of the error amplifier. Its voltage is the control voltage that sets the duty cycle. A network here stabilizes a feedback loop.
10	SHUTDOWN	When pulled high, the outputs turn off and stay off until the next clock pulse after it is released. Used for protection.
11	OUT A	Output A of the totem-pole stage. Switches at half the oscillator frequency.
12	GND	Ground reference for everything.
13	VC	Supply for the output stage only. It sets how high the gate-drive voltage is.
14	OUT B	Output B of the totem-pole stage, 180 degrees out of phase with A.
15	VIN	Supply for everything inside the IC except the output transistors.
16	VREF	The precise 5.1 V reference. It powers the internal analog circuits and sets the accuracy of the output regulation. A bypass capacitor is connected here.

4. What sets the frequency (cause and effect)

The oscillator frequency depends on three components:

$$f_{osc} = 1 / (Ct \times (0.7 \times Rt + 3 \times Rd))$$

and each output switches at $f_{out} = f_{osc} / 2$.

If you change	Effect on the sawtooth	Effect on frequency
Rt up	Ct charges more slowly, so the ramp takes longer	Frequency goes down
Rt down	Ct charges faster	Frequency goes up
Ct up	A bigger capacitor takes longer to reach the top	Frequency goes down
Ct down	Reaches the top sooner	Frequency goes up
Rd up	Discharge takes longer, so each cycle is slightly longer	Frequency drops slightly, dead time grows

The frequency does **not** depend on the control voltage, the supply voltage (within the allowed range), or the duty cycle. Changing the duty cycle leaves the frequency unchanged.

Example: Rt = 7.4 kΩ, Ct = 10 nF, Rd = 10 Ω gives f_osc of about 20 kHz and **f_out of about 10 kHz** per output. In the LTspice model the measured value was within a few percent of the formula, since the formula is an approximation.

5. What sets the duty cycle (cause and effect)

The duty cycle is set by **where the control voltage on CMP crosses the sawtooth**:

- While the sawtooth is below the control voltage, the output is high.
- When the sawtooth rises above it, the output goes low.

Control voltage (CMP)	Crossing point on the ramp	Result
At or below about 0.9 V	At the very start	Duty is about 0% (no pulse)
Around the middle (about 2.1 V)	Halfway up the ramp	Medium duty
At or above about 3.3 V	At the very end	Maximum duty (about 49%)

Approximate relation (for the middle region):

$$\textbf{Duty} \approx 49\% \times (V_{CMP} - 0.9\ V) / (3.3\ V - 0.9\ V)$$

And the control voltage itself comes from the error amplifier:

$V_{CMP} = A \times (V_{NI} - V_{INV})$, where A is the open-loop gain, very large.

6. Why the duty cycle only reaches about 49% at each output

A single output cannot reach 100% duty for two reasons that work together:

1. **The outputs take turns.** The oscillator runs at f_osc, but the flip-flop sends the pulses to OUA and OUB alternately. Each output is therefore allowed to be on only during its own half of the cycle. Its period is two oscillator periods, but it can only use one oscillator period of that time, so its limit is **50%**.
2. **Dead time removes a little more.** When Ct is discharged through Rd, a short clock pulse forces both outputs off. This gap prevents both outputs being on together. It is about 1 to 2 µs in our simulation. With T_osc = 50 µs, this loses roughly 1 to 2 percentage points, leaving about **49%**.

$$\textbf{Maximum duty per output} \approx 0.5 \times (1 - t_{dead} / T_{osc})$$

A larger Rd (or Ct) lengthens the dead time and lowers the maximum duty. The dead time is also useful in practice: In a forward converter it gives the transformer core time to reset before the next pulse.

7. Quick reference

Quantity	Formula or value
Reference voltage (pin 16)	5.1 V
Sawtooth range (pin 5)	about 0.9 V to 3.3 V
Oscillator frequency	f_osc = 1 / (Ct × (0.7 Rt + 3 Rd))
Output frequency per pin	f_out = f_osc / 2
Duty (middle region)	≈ 49% × (V_CMP − 0.9) / 2.4
Maximum duty per output	≈ 0.5 × (1 − t_dead / T_osc), about 49%
Error amplifier	V_CMP = A × (V_NI − V_INV)
Follower (CMP tied to INV)	V_CMP = V_NI × A / (1 + A) ≈ V_NI

8. Output voltage and current (from the datasheet)

Values below are from the STMicroelectronics SG2525A/SG3525A datasheet (June 2000). The output stage is a totem-pole driver (one transistor pulls the output up, one pulls it down) with NOR logic, so an OFF state gives a LOW output. It can source or sink more than 200 mA.

Parameter	Test condition	Typical value	Limit
Output high level (amplitude)	Isource = 20 mA, VC = 20 V	19 V	18 V min
Output high level (amplitude)	Isource = 100 mA, VC = 20 V	18 V	17 V min
Output low level	Isink = 20 mA	0.2 V	0.4 V max
Output low level	Isink = 100 mA	1 V	2 V max

Parameter	Test condition	Typical value	Limit
Output current, steady state	Source or sink	up to 100 mA	recommended operating range
Output current, peak	Source or sink	up to 400 mA	recommended operating range
Output current, absolute maximum	Source or sink	not applicable	500 mA
Collector (output stage) supply VC	Recommended range	4.5 V to 35 V	40 V absolute maximum
Rise time	CL = 1 nF	100 ns	600 ns max
Fall time	CL = 1 nF	50 ns	300 ns max
Under-voltage lockout	Outputs stay off below this supply	7 V	6 V to 8 V

What this means in practice:

- **The output swings from about 0 V to a little below VC.** The high level is VC minus the voltage drop in the top transistor, about 1 V at 20 mA and about 2 V at 100 mA. The low level is close to 0 V (0.2 V at 20 mA).
- **The high level is set by VC (pin 13), not by VIN (pin 15).** To drive a MOSFET gate, VC is chosen to give the required gate voltage.
- **Example from our simulation:** with VC = 15 V and a 1 kΩ load, the load current is about 14 V / 1 kΩ = 14 mA. This is well below the 100 mA steady-state limit, and the output measured about 14 V, in line with the datasheet drop.
- **Output current is only needed in short bursts when driving a MOSFET.** The gate capacitance is charged and discharged on each edge, so peak current (up to 400 mA recommended) matters more than average current.
- **Rise and fall times are short** (100 ns and 50 ns typical with 1 nF), so the outputs switch fast compared with a 10 kHz period of 100 µs.